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8/30

ANSWER BOOK

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Subject Design & Analysis of Algorithms Semester 3

Mark Tabulation

Mark Divisions	Q.1	Q.2	Q.3	Q.4	Q.5	Q.6	Q.7	Q.8	Q.9	Q.10	Q.11	Q.12	Q.13	Q.14	Q.15
		1	1/2	1/2	3	1	1	0	3						
Total													9		

Grand Total

5. Minimum spanning tree is the shortest distance a tree can have including all its ed vertices. There are various algorithms to find th minimum spanning tree, they are prims algori kruskals algorithm etc. ~~Spanning~~ Minimum span ~~for example,~~ tree must not be a close tree.



For example,

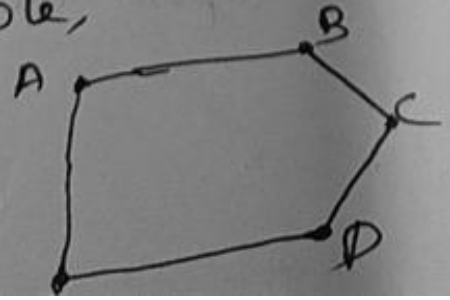


Fig (A)

This figure can be changed to minimum spanning tree by



Minimum spanning tree of fig (A)

This includes all the vertices of the fig (A) and does not have a closed figure.

2. Time complexity of an algorithm quantifies the amount of time taken by an algorithm to run as a function of the length of the input. Similarly, space complexity of an algorithm quantifies the amount of space or memory taken by an algorithm to run as a function of the length of the input. Time and space complexity depends on lots of things like hardware, operating system, processors etc. However, we don't consider any of these factors while analyzing the algorithm. Order of growth is how the time of execution depends on the length of the input. Order of growth will help us to compute the running time

Profit
Weight of $P_1 = \frac{500}{1} = 500$

Profit
Weight of $P_2 = \frac{900}{3} = 300$

Profit
Weight of $P_3 = \frac{700}{7} = 100$

Knapsack capacity = 5

∴ Arranged Profit

$= P_1 \; P_3 \; P_2$

∴ Weight = 1 4/7 0

∴ Profit = 500 + 2800/7

500 400



6. In job sequencing problem, the objective is to find a sequence of jobs, which is completed with their deadlines and gives maximum profit.

Algorithm (Job sequencing ~~problem~~ with Deadline (D, J, n, k)

D(0) := J(0) := 0

k := 1

J(1) = 1

for i = 2 - n do

r := k

while D(J(r)) > D(i) and D(J(r)

r := r - 1

if D(J(r)) ≤ D(i) and D

then

for l = k - r + 1 by -1 do

J(l + 1) := J(l)

J(r+1) := j

k := k+1

The sequencing of jobs on a single processor with deadline constraints is called as Job sequencing with Deadlines. Each job has a defined deadlines and some profit associated with it. The profit of a job is given only when that job is completed within its deadline.

7. Greedy algorithm is optimization method. When the problem has many feasible solutions with different cost or benefit, finding the best solution is known as an optimization problem and the best solution is known optimal solution.

2 Control abstraction for Greedy Method.

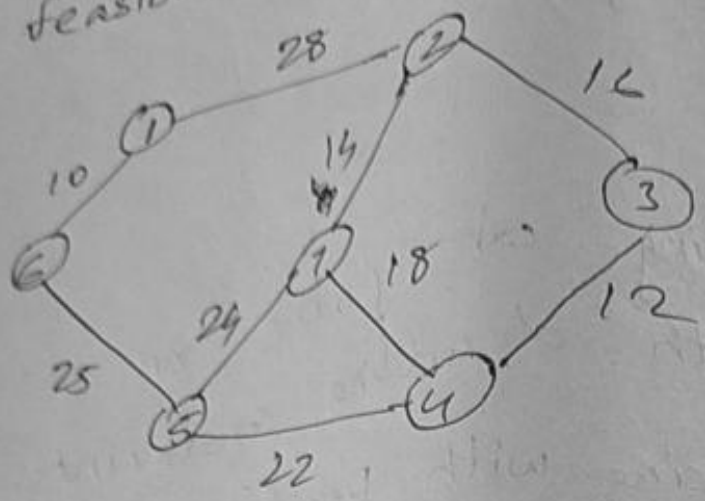
. A selection of solution from the given in domain is performed, ie.., s = select(a).

. The feasibility of a solution is perfor

Final solution. This is the spanning

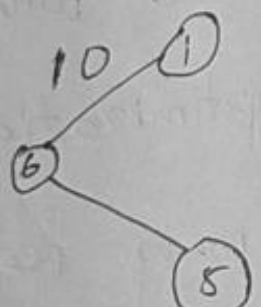
by using ... feasible solu...

9.



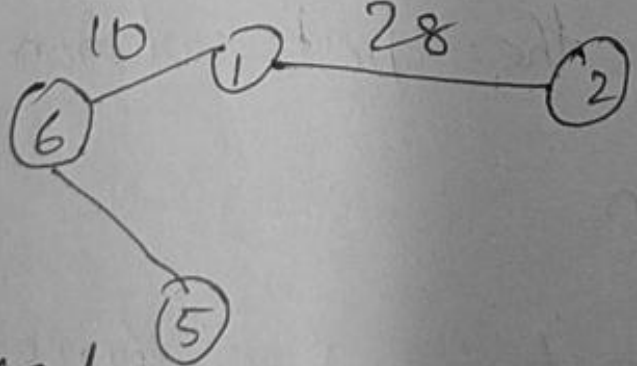
Consider 6 as the vertex.

mstSet = { 6, INF, INF, INF, INF, INF, IN



Shortest distance 1 is sel

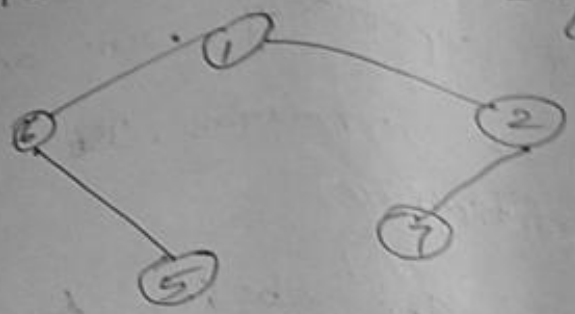
mstSet = { 6, 1, INF, INF, INF, INF, 1



Adjacent vertex 2 is selected.

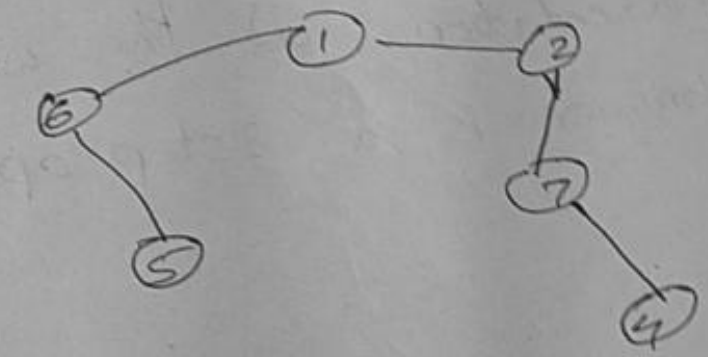
mstSet = { 6, 1, INF, INF, INF, 2

is 7. mstset = { 6, 1, [illegible] 2, 7, INF, INF }

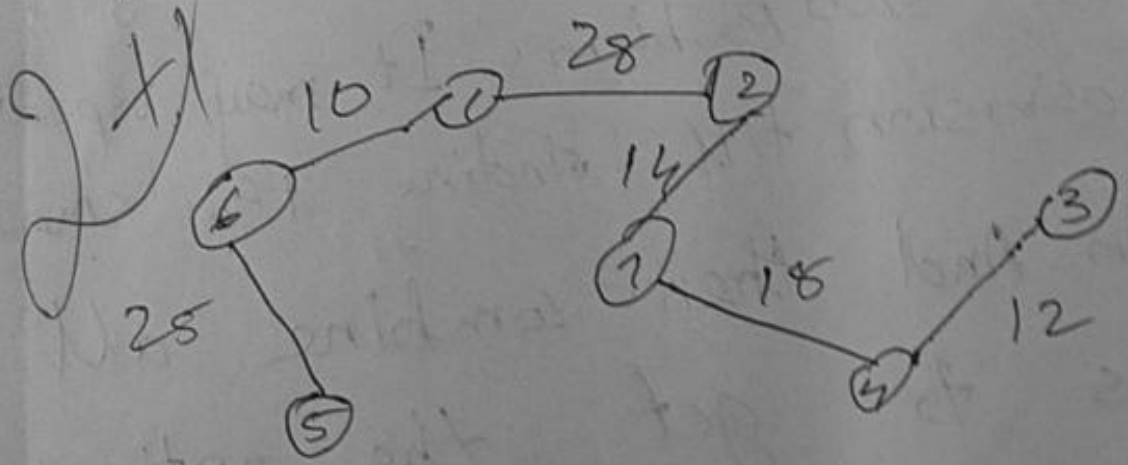


Adjacent vertex of 7 is 5 and 4. we select 4.

mstSet: { 6, 1, 2, 7, 4, INF, INF }



Adjacent vertices of 4 are 3 and 5 we select 3.



mstSet: { 6, 1, 2, 7, 4, 3, INF }

using prime algorithms

3. Control abstraction is the process by u
programmers define new control constructs, specify
a statement ordering seperately from an imp
ntantion of that ordering.

Two types of abstraction :-

Control abstraction : The ability to procedural data where in the program procedural data is irrelevant.

Data abstraction :- It is the reduction of a particular body of data to a simplified representation of the whole.

Divide and conquer is a 3step approach.

- Divide
- Conquer
- Combine.

1/2

In first step, divide and conquer approac make algorithm to divide the big problem into small sub problems. It may repeatedly do this division till finding the smalles subproblem. And then combine all subproblems to get the optimal so